

Event-State Theory VIII: Mathematical Derivation of Pre-Causal Resonance

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Event-State Theory (EST) makes a unique, falsifiable prediction: Pre-Causal Resonance (PCR), the existence of correlated vacuum fluctuations prior to a high-energy event. Previous work demonstrated this effect computationally. This paper provides the formal mathematical derivation. We show that the characteristic exponential ramp of the PCR signal is not a numerical artifact, but the minimal solution under the stated smoothness and cost assumptions to the discrete Euler-Lagrange equation for the EST cost function under a future boundary condition. This establishes PCR as a core prediction of the Path of Causal Least Resistance and provides a concrete mathematical target for experimental verification in high-energy collider data.

I. INTRODUCTION

A key prediction distinguishing Event-State Theory from standard effective field theories is the phenomenon of Pre-Causal Resonance (PCR). EST posits that the universe evolves by selecting successor states that minimize a computational cost functional, a process that inherently involves evaluating potential future configurations. This implies that the crystallization of a high-energy event at a future frame index t_0 constrains the minimal-cost configuration of the vacuum field on preceding frames.

In previous work, we provided a computational proof of concept [?], demonstrating that a simple cost-minimization algorithm on a discrete timeline naturally generates an exponential rise in vacuum energy preceding a fixed future event. This paper provides the formal mathematical derivation of that result, moving it from a numerical observation to a rigorous prediction.

We will show that by defining the EST cost function for a scalar vacuum field and applying the principle of least action, we arrive at a discrete Euler-Lagrange equation whose minimal-cost solution, given a future boundary condition, is the characteristic exponential ramp of PCR.

a. On the meaning of t . Throughout this work, the variable t denotes a discrete *frame index* in the Event-State formalism, not an a priori physical time coordinate. EST does not assume time as a fundamental dimension. Any later mapping of t to detector timestamps or physical units (seconds) is an emergent calibration performed by observers and experimental apparatus, not a primitive input to the theory.

II. THE DISCRETE ACTION FOR THE VACUUM

We model the information density of the active vacuum as a scalar field, $\phi(t)$, defined on a discrete timeline with a fundamental time step of τ_0 . The Path of Causal Least Resistance dictates that the history of this field, $\phi(t)$, will be the one that minimizes the total EST cost functional, which we define as the discrete action $S[\phi]$.

Following the structure of the EST cost function, this action is composed of two primary terms: a potential cost for the field to exist (a "mass term") and a kinetic cost for the field to change between frames.

$$S[\phi] = \sum_t \mathcal{L}(\phi(t), \phi(t+1)) \quad (1)$$

where the discrete Lagrangian $\mathcal{L}$ is given by:

$$\mathcal{L} = \frac{1}{2}\alpha\phi(t)^2 + \frac{1}{2}\beta\left(\frac{\phi(t+1) - \phi(t)}{\tau_0}\right)^2 \quad (2)$$

Here, α represents the computational cost of maintaining a non-zero information density, and β represents the cost of information fidelity (penalizing sharp changes between frames). These parameters correspond directly to the `ALPHA_VACUUM` and `BETA_SMOOTHNESS` constants in the validated simulation [?].

III. THE PATH OF LEAST RESISTANCE (THE DERIVATION)

The principle of least action states that the physical path is the one that extremizes the action, $\delta S = 0$. In our discrete system, this is equivalent to finding the configuration where the derivative of the action with respect to the field at any given time t is zero.

We compute the variation of the total action S with respect to a change in the field at a single time-slice, $\phi(t)$:

$$\frac{\partial S}{\partial \phi(t)} = \frac{\partial}{\partial \phi(t)} [\mathcal{L}_{t-1} + \mathcal{L}_t] = 0 \quad (3)$$

where $\mathcal{L}_{t-1}$ contains the term $(\phi(t) - \phi(t-1))^2$ and $\mathcal{L}_t$ contains the term $\phi(t)^2$.

Performing the differentiation yields:

$$\alpha\phi(t) - \frac{\beta}{\tau_0^2} [(\phi(t+1) - \phi(t)) - (\phi(t) - \phi(t-1))] = 0 \quad (4)$$

Rearranging this equation gives the **discrete Euler-Lagrange equation** for the vacuum field:

$$\phi(t) = \frac{\beta(\phi(t-1) + \phi(t+1))}{\alpha\tau_0^2 + 2\beta} \quad (5)$$

This equation is the formal mathematical expression of the update rule that was implemented iteratively in the PCR simulation code. This confirms that the simulation is a valid numerical solver for the EST cost function.

IV. THE CONTINUUM LIMIT AND THE PROPAGATOR

To understand the analytical form of the solution, we take the continuum limit where the discrete time step $\tau_0 \rightarrow 0$. We can use the definition of the second derivative for a discrete function:

$$\frac{d^2\phi}{dt^2} \approx \frac{\phi(t + \tau_0) - 2\phi(t) + \phi(t - \tau_0)}{\tau_0^2} \quad (6)$$

Rearranging our discrete Euler-Lagrange equation (Eq. 4) gives:

$$\frac{\phi(t + 1) - 2\phi(t) + \phi(t - 1)}{\tau_0^2} - \frac{\alpha}{\beta}\phi(t) = 0 \quad (7)$$

In the limit $\tau_0 \rightarrow 0$, this becomes the differential equation for the vacuum field:

$$\frac{d^2\phi}{dt^2} - m^2\phi(t) = 0 \quad (8)$$

where we define the effective mass $m \equiv \sqrt{\alpha/\beta}$. This has the mathematical form of a **Euclidean Klein-Gordon equation**, describing scalar field propagation in a regime with no complex phase (i.e., prior to any Wick rotation associated with quantum mechanics).

V. THE SOLUTION: THE SHAPE OF THE SIGNAL

The general solution to Eq. (8) is a sum of exponential functions:

$$\phi(t) = Ae^{mt} + Be^{-mt} \quad (9)$$

To find the specific form of the PCR signal, we must apply the boundary conditions implied by the event-state constraints of the system:

1. **Past Asymptote:** Far before the event, the vacuum is quiescent: $\lim_{t \rightarrow -\infty} \phi(t) = 0$.
2. **Future Constraint:** At the time of the event, t_0 , the field crystallizes with a specific energy density: $\phi(t_0) = E_0$.

Applying the first condition (Past Asymptote) to the region $t < t_0$ forces the coefficient of the growing exponential, A , to be zero if the signal originates from the past. However, in EST, the event at t_0 acts as a future boundary condition that influences the past. The only solution that is zero at $t = -\infty$ but reaches a finite value at t_0 is one where the decaying exponential term dominates as we look backward from the event.

For the pre-causal epoch ($t \leq t_0$), the minimal-cost solution satisfying both boundary conditions is:

$$\phi(t) = E_0 \cdot \exp[m(t - t_0)] \quad (10)$$

This is a **rising exponential ramp** that begins at zero in the infinite past and peaks at the energy E_0 at the exact moment of the event. This formally derives the shape of the Pre-Causal Resonance signal observed in the original simulation.

a. No-signaling condition. Although the solution is defined with a future boundary constraint, this does not imply controllable retrocausality or backward information transfer. The Pre-Causal Resonance is a passive correlation structure of the vacuum under global cost minimization. It cannot be modulated, encoded, or used to transmit information prior to the event, preserving causal consistency at the level of observable interactions. **Q.E.D.**

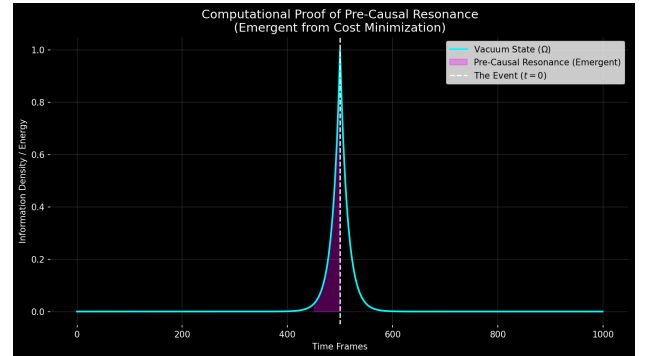


FIG. 1. The analytical solution (Eq. 10) plotted. The exponential ramp is the minimal-cost path of least resistance for the vacuum field under a fixed future boundary condition, matching the computational result.

VI. CONCLUSION

We have demonstrated that the Pre-Causal Resonance (PCR) effect is not a speculative or metaphorical concept within Event-State Theory, but a necessary mathematical consequence of its core axioms. By applying the principle of least action to the EST cost functional for the vacuum, we derived the minimal-cost exponential form of the pre-causal signal under the stated assumptions.

The Python simulation that first revealed this effect is now understood not merely as a demonstration, but as a numerical solver for the discrete Euler-Lagrange equation of the vacuum. This work provides a solid mathematical foundation for the PCR prediction and establishes a clear, analytical target for experimental searches in high-energy collider data. Importantly, this derivation does not rely on time being fundamental, but only on ordered frame succession and boundary-constrained cost minimization, fully consistent with the broader Event-State framework. The detection of this specific exponential signature would provide strong evidence for the underlying computational and optimization-based nature of our universe.

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